Homework Assignment

Singular Value Decomposition

Theory, Algorithms, and Applications

Course Topic: Singular Value Decomposition, Pseudoinverse, Low-Rank Approximation, and Applications

Prepared as a companion assignment for the SVD lecture notes

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Contents

Instructions 2

Problem 1. Singular Values and Singular Vectors 2

Problem 2. Existence of the Singular Value Decomposition 2

Problem 3. Compact SVD and Outer Product Expansion 3

Problem 4. Fundamental Subspaces and the SVD 3

Problem 5. Matrix Norms and Singular Values 4

Problem 6. Moore-Penrose Pseudoinverse 4

Problem 7. Least-Squares Problems and the Pseudoinverse 5

Problem 8. Rank-Deficient Least-Squares Problems 5

Problem 9. Eckart-Young-Mirsky Theorem 6

Problem 10. Low-Rank Approximation and Data Compression 7

Problem 11. Principal Component Analysis 7

Problem 12. Numerical Experiment: Image Compression by SVD 8

Problem 13. Numerical Experiment: Low-Rank Approximation of a Data Matrix 8

Problem 14. Regularization and Small Singular Values 9

Problem 15. Short Essay Question 10

Optional Challenge Problem 10

Submission Requirements 10

Instructions

Answer all problems carefully. For theoretical questions, provide complete derivations or proofs. For computational problems, include your code, numerical results, and a short discussion. You may use MATLAB, Python, Julia, or another scientific computing language.

Problem 1. Singular Values and Singular Vectors

Let $A \in \mathbb{R}^{m\times n}$. The singular values of A are defined as the nonnegative square roots of the eigenvalues of A^TA.

Tasks.

1. Prove that A^TA is symmetric positive semidefinite.
2. Show that all eigenvalues of A^TA are nonnegative.
3. Let
$$A^TAv_i = \sigma_i^2 v_i,$$
where $v_i \neq 0$. Prove that
$$\|Av_i\|_2 = \sigma_i \|v_i\|_2.$$
4. Suppose v_i is normalized and $\sigma_i > 0$. Define
$$u_i = \frac{Av_i}{\sigma_i}.$$
Prove that u_i is a unit vector.
5. Explain the geometric meaning of singular values as stretching factors of a linear transformation.

Problem 2. Existence of the Singular Value Decomposition

Let $A \in \mathbb{R}^{m\times n}$. The singular value decomposition of A is
$$A = U\Sigma V^T,$$
where $U \in \mathbb{R}^{m\times m}$ and $V \in \mathbb{R}^{n\times n}$ are orthogonal matrices, and $\Sigma \in \mathbb{R}^{m\times n}$ is diagonal in the rectangular sense.

Tasks.

1. Starting from the spectral decomposition of A^TA, derive the SVD of A.
2. Explain why the columns of V can be chosen as an orthonormal eigenbasis of A^TA.
3. For each positive singular value σ_i, define
$$u_i = \frac{Av_i}{\sigma_i}.$$
Prove that the vectors u_i are orthonormal.
4. Explain how to complete the set $\{u_1, \ldots, u_r\}$ to an orthonormal basis of $\mathbb{R}^m$, where $r = \operatorname{rank}(A)$.
5. Conclude that every real matrix admits a singular value decomposition.

Problem 3. Compact SVD and Outer Product Expansion

Let $A \in \mathbb{R}^{m\times n}$ have rank r. Its compact SVD is

$$A = U_r \Sigma_r V_r^T,$$

where

$$U_r = [u_1, \dots, u_r], \qquad V_r = [v_1, \dots, v_r],$$

and

$$\Sigma_r = \operatorname{diag}(\sigma_1, \dots, \sigma_r),$$

with

$$\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r > 0.$$

Tasks.

1. Prove that the compact SVD can be written as the outer product expansion
$$A = \sum_{i=1}^{r} \sigma_i u_i v_i^T.$$
2. Show that each matrix
$$u_i v_i^T$$
has rank one.
3. Explain why the SVD represents A as a sum of mutually orthogonal rank-one components.
4. Compute the compact SVD of
$$A = \begin{bmatrix} 3 & 0 \\ 0 & 2 \\ 0 & 0 \end{bmatrix}.$$
5. Write the corresponding outer product expansion.

Problem 4. Fundamental Subspaces and the SVD

Let

$$A = U\Sigma V^T$$

be the SVD of $A \in \mathbb{R}^{m\times n}$, and suppose $\operatorname{rank}(A) = r$.

Tasks.

1. Prove that
$$\mathcal{R}(A) = \operatorname{span}\{u_1, \dots, u_r\}.$$
2. Prove that
$$\mathcal{N}(A) = \operatorname{span}\{v_{r+1}, \dots, v_n\}.$$
3. Prove that
$$\mathcal{R}(A^T) = \operatorname{span}\{v_1, \dots, v_r\}.$$
4. Prove that
$$\mathcal{N}(A^T) = \operatorname{span}\{u_{r+1}, \dots, u_m\}.$$
5. Explain how the SVD gives orthonormal bases for all four fundamental subspaces of a matrix.

Problem 5. Matrix Norms and Singular Values

Let $A \in \mathbb{R}^{m\times n}$ have singular values

$$\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r > 0.$$

Tasks.

1. Prove that the spectral norm of A is
$$\|A\|_2 = \sigma_1.$$
2. Prove that the Frobenius norm of A is
$$\|A\|_F = \left(\sum_{i=1}^{r} \sigma_i^2\right)^{1/2}.$$
3. Prove that
$$\|A\|_2^2 = \lambda_{\max}(A^T A).$$
4. Show that if A is square and nonsingular, then
$$\|A^{-1}\|_2 = \frac{1}{\sigma_n}.$$
5. Deduce that the spectral condition number of a nonsingular square matrix is
$$\kappa_2(A) = \frac{\sigma_1}{\sigma_n}.$$

Problem 6. Moore-Penrose Pseudoinverse

Let
$$A = U\Sigma V^T$$
be the SVD of $A \in \mathbb{R}^{m\times n}$. The Moore-Penrose pseudoinverse is defined by
$$A^+ = V\Sigma^+ U^T,$$
where Σ^+ is obtained by replacing each positive singular value σ_i by $1/\sigma_i$ and transposing the rectangular diagonal matrix.

Tasks.

1. Write down Σ^+ explicitly for the case $m > n$ and $\operatorname{rank}(A) = r$.
2. Prove the four Moore-Penrose conditions:
$$AA^+A = A,$$
$$A^+AA^+ = A^+,$$
$$(AA^+)^T = AA^+,$$
$$(A^+A)^T = A^+A.$$

3. Show that if A is square and nonsingular, then
$$A^+ = A^{-1}.$$

4. Compute the pseudoinverse of
$$A = \begin{bmatrix} 1 & 0 \\ 0 & 2 \\ 0 & 0 \end{bmatrix}.$$

5. Verify directly that your result satisfies the Moore-Penrose conditions.

Problem 7. Least-Squares Problems and the Pseudoinverse

Consider the least-squares problem
$$\min_{x \in \mathbb{R}^n} \|Ax - b\|_2,$$
where $A \in \mathbb{R}^{m \times n}$.

Tasks.

1. Derive the normal equations
$$A^T A x = A^T b.$$

2. Suppose A has full column rank. Prove that the unique least-squares solution is
$$x = (A^T A)^{-1} A^T b.$$

3. Using the SVD, show that the minimum-norm least-squares solution is
$$x = A^+ b.$$

4. Let
$$A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}, \qquad b = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}.$$
Compute the least-squares solution.

5. Compute the residual vector
$$r = b - Ax.$$

6. Verify that
$$A^T r = 0.$$

Problem 8. Rank-Deficient Least-Squares Problems

Let $A \in \mathbb{R}^{m \times n}$ be rank deficient. Consider
$$\min_x \|Ax - b\|_2.$$

Tasks.

1. Explain why the normal equations may have infinitely many solutions.
2. Prove that among all least-squares solutions, A^+b has minimum Euclidean norm.
3. Consider
$$A = \begin{bmatrix} 1 & 1 \\ 2 & 2 \\ 3 & 3 \end{bmatrix}, \qquad b = \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix}.$$
Find the minimum-norm least-squares solution.
4. Describe geometrically what happens when the columns of A are linearly dependent.
5. Explain why the SVD is more reliable than solving the normal equations for rank-deficient problems.

Problem 9. Eckart-Young-Mirsky Theorem

Let
$$A = \sum_{i=1}^{r} \sigma_i u_i v_i^T$$
be the compact SVD of A, where
$$\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r > 0.$$
Define the truncated SVD approximation
$$A_k = \sum_{i=1}^{k} \sigma_i u_i v_i^T, \qquad k < r.$$

Tasks.

1. Prove that
$$\operatorname{rank}(A_k) = k.$$
2. State the Eckart-Young-Mirsky theorem for the spectral norm.
3. State the Eckart-Young-Mirsky theorem for the Frobenius norm.
4. Prove that
$$\|A - A_k\|_2 = \sigma_{k+1}.$$
5. Prove that
$$\|A - A_k\|_F = \left(\sum_{i=k+1}^{r} \sigma_i^2 \right)^{1/2}.$$
6. Explain why A_k is the best rank-k approximation to A.

Problem 10. Low-Rank Approximation and Data Compression

Let $A \in \mathbb{R}^{m \times n}$ be a data matrix or grayscale image matrix. Suppose the singular values of A are

$$\sigma_1, \sigma_2, \ldots, \sigma_r.$$

Tasks.

1. Explain how the truncated SVD
$$A_k = U_k \Sigma_k V_k^T$$
can be used to compress the matrix A.
2. Count the number of scalar values required to store A directly.
3. Count the number of scalar values required to store A_k.
4. Derive the condition under which storing A_k requires fewer numbers than storing A.
5. Define the relative Frobenius approximation error
$$E_k = \frac{\|A - A_k\|_F}{\|A\|_F}.$$
6. Express E_k in terms of the singular values of A.
7. Explain how the decay rate of singular values affects the effectiveness of compression.

Problem 11. Principal Component Analysis

Let $X \in \mathbb{R}^{m \times n}$ be a centered data matrix, where each row represents an observation and each column represents a variable. The sample covariance matrix is

$$C = \frac{1}{m-1} X^T X.$$

Tasks.

1. Let
$$X = U \Sigma V^T$$
be the SVD of X. Show that
$$C = V \left(\frac{\Sigma^T \Sigma}{m-1} \right) V^T.$$
2. Explain why the right singular vectors of X are the principal directions.
3. Show that the variance explained by the i-th principal component is
$$\frac{\sigma_i^2}{m-1}.$$
4. Derive the cumulative explained variance ratio
$$R_k = \frac{\sum_{i=1}^{k} \sigma_i^2}{\sum_{i=1}^{r} \sigma_i^2}.$$
5. Describe how to choose k for dimensionality reduction using R_k.
6. Give one practical example where PCA can be useful.

Problem 12. Numerical Experiment: Image Compression by SVD

Choose a grayscale image and represent it as a matrix

$$A \in \mathbb{R}^{m \times n}.$$

Tasks.

1. Compute the singular value decomposition of A.
2. Construct rank-k approximations
$$A_k = U_k \Sigma_k V_k^T$$
for
$$k = 5,\ 10,\ 20,\ 50,\ 100.$$
3. For each k, compute the relative Frobenius error
$$E_k = \frac{\|A - A_k\|_F}{\|A\|_F}.$$
4. For each k, compute the compression ratio
$$\rho_k = \frac{k(m+n+1)}{mn}.$$
5. Display the reconstructed images for different k.
6. Plot the singular values σ_i on a semilog scale.
7. Discuss the relationship between singular value decay and image quality.

Problem 13. Numerical Experiment: Low-Rank Approximation of a Data Matrix

Generate a matrix
$$A \in \mathbb{R}^{100 \times 80}$$
of the form
$$A = B + C,$$
where B is a rank-5 matrix and C is a small noise matrix.

Tasks.

1. Construct B as
$$B = PQ^T,$$
where
$$P \in \mathbb{R}^{100 \times 5}, \qquad Q \in \mathbb{R}^{80 \times 5}.$$
2. Add random noise
$$C = \varepsilon N,$$
where N has independent standard normal entries and $\varepsilon > 0$ is small.

3. Compute the singular values of A.
4. Plot the singular values.
5. Explain how the singular values reveal the approximate rank of A.
6. Compute rank-k approximations for
$$k = 1, 2, \ldots, 20.$$
7. Plot the relative approximation error versus k.
8. Discuss why the error decreases rapidly near $k = 5$.

Problem 14. Regularization and Small Singular Values

Consider an ill-conditioned least-squares problem
$$\min_x \|Ax - b\|_2,$$
where A has singular values satisfying
$$\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r > 0,$$
and some singular values are very small.

Tasks.

1. Using the SVD, show that the pseudoinverse solution can be written as
$$x = A^+ b = \sum_{i=1}^{r} \frac{u_i^T b}{\sigma_i} v_i.$$
2. Explain why small singular values may amplify noise in b.
3. Define the truncated SVD regularized solution
$$x_k = \sum_{i=1}^{k} \frac{u_i^T b}{\sigma_i} v_i.$$
4. Explain how truncated SVD reduces noise amplification.
5. Compare truncated SVD regularization with Tikhonov regularization, whose solution has the form
$$x_\lambda = \sum_{i=1}^{r} \frac{\sigma_i}{\sigma_i^2 + \lambda} (u_i^T b) v_i.$$
6. Discuss the role of the regularization parameter λ.

Problem 15. Short Essay Question

Write a short essay of about one page on the following topic:

"Why Singular Value Decomposition Is a Fundamental Tool in Numerical Linear Algebra"

Your essay should address the following points:

1. The geometric interpretation of SVD.
2. The role of singular values in measuring rank, conditioning, and stability.
3. The use of SVD in least-squares problems and pseudoinverses.
4. The optimality of truncated SVD for low-rank approximation.
5. Applications in data science, image compression, PCA, and regularization.

Optional Challenge Problem

Let $A \in \mathbb{C}^{m\times n}$. The complex singular value decomposition is

$$A = U\Sigma V^*,$$

where U and V are unitary matrices.

Tasks.

1. Prove that A^*A is Hermitian positive semidefinite.
2. Derive the complex SVD from the spectral theorem for Hermitian matrices.
3. Prove that the singular values of A and A^* are the same.
4. Prove that
$$\|A\|_2 = \sigma_1.$$
5. Explain the difference between the transpose V^T in the real SVD and the conjugate transpose V^* in the complex SVD.

Submission Requirements

Your submission should include:

1. Complete written solutions to all theoretical problems.
2. Source code for numerical experiments.
3. Tables of numerical errors and compression ratios.
4. Plots of singular values and approximation errors.

5. Reconstructed images or data visualizations where appropriate.

6. A short discussion of numerical observations.

Recommended format: PDF report with code included in an appendix.